

ZONGULDAK, Turkey

WMO: 170220

Lat: 41.4495N

Lon: 31.7785E

Elev: 449

StdP: 14.46

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.6	31.8	19.7	15.0	38.4	22.5	17.2	38.7	19.7	47.7	17.3	47.3	6.8	180	0.360

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.2	83.3	69.9	81.2	70.9	79.8	70.5	75.4	80.2	74.0	79.0	72.7	77.8	5.7	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.8	127.9	78.8	72.2	121.2	77.8	70.8	115.5	76.8	39.2	80.3	37.9	79.2	36.7	78.0	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.4	13.0	11.2	27.4	90.8	3.1	4.4	25.2	93.9	23.3	96.5	21.6	98.9	19.3	102.1		
(5)			25.6	77.0	3.1	2.2	23.4	78.6	21.6	80.0	19.9	81.2	17.6	82.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	57.7	43.8	44.3	47.2	53.5	60.8	68.0	72.5	73.0	67.4	60.2	53.3	47.1	(6)
(7)		DBStd	12.25	7.68	7.72	8.31	8.01	6.41	4.46	3.19	3.15	4.54	6.10	7.02	7.84	(7)
(8)		HDD50	825	226	194	160	47	1	0	0	0	0	4	42	151	(8)
(9)		HDD65	3446	659	580	554	360	165	18	0	0	26	173	355	556	(9)
(10)		CDD50	3624	32	34	74	151	336	540	699	712	523	319	142	62	(10)
(11)		CDD65	770	0	0	2	14	35	108	234	247	100	23	5	2	(11)
(12)		CDH74	3353	0	1	14	77	159	383	1102	1226	301	70	12	8	(12)
(13)		CDH80	367	0	0	2	18	27	59	107	104	39	11	0	0	(13)
(14)	Wind	WSAvg	4.7	5.2	5.1	5.1	4.4	4.2	4.3	4.6	4.7	4.8	4.6	4.7	5.3	(14)
(15)	Precipitation	PrecAvg	47.1	5.2	3.6	3.7	2.3	2.3	2.8	3.0	3.3	4.3	5.6	5.6	6.3	(15)
(16)		PrecMax	65.6	9.8	8.1	7.1	6.1	9.1	9.0	10.3	13.4	13.9	12.8	14.2	10.9	(16)
(17)		PrecMin	32.4	0.2	0.6	0.7	0.2	0.4	0.7	0.2	0.0	0.0	0.6	0.4	1.6	(17)
(18)		PrecStd	6.6	2.3	1.7	1.6	1.3	1.5	1.8	2.7	3.1	3.4	2.7	3.1	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.3	69.0	75.2	82.6	83.1	86.3	87.6	85.5	84.2	80.0	75.7	73.8	(19)
(20)			MCWB	50.5	53.1	55.2	59.0	63.1	67.6	69.9	69.6	67.0	63.0	58.0	53.8	(20)
(21)		2%	DB	61.9	63.5	68.2	75.0	78.4	80.6	82.3	82.3	79.9	75.2	70.0	66.6	(21)
(22)			MCWB	49.1	50.0	53.2	57.8	62.6	67.7	72.7	74.1	68.0	62.0	56.1	52.8	(22)
(23)		5%	DB	58.9	59.5	63.9	69.9	74.3	78.0	80.7	80.8	77.0	71.5	66.6	62.6	(23)
(24)			MCWB	47.8	48.4	51.0	55.9	61.1	67.6	72.5	73.0	68.1	62.9	54.8	50.8	(24)
(25)		10%	DB	55.6	56.2	60.3	65.4	70.7	75.6	79.1	79.5	75.1	68.6	63.6	58.9	(25)
(26)			MCWB	46.4	46.9	49.2	53.5	60.1	67.3	71.4	72.1	67.0	62.0	53.9	49.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.4	54.5	57.5	61.4	66.4	73.8	76.6	78.5	73.0	67.8	61.0	56.4	(27)
(28)			MCD B	62.7	64.4	71.2	77.5	75.8	78.8	81.0	81.4	78.2	72.1	67.3	67.3	(28)
(29)		2%	WB	50.3	51.4	54.2	58.5	64.5	71.6	75.1	76.2	71.1	65.8	58.7	53.9	(29)
(30)			MCD B	59.0	60.5	65.4	71.7	72.7	76.3	80.3	80.3	76.2	70.4	66.7	63.6	(30)
(31)		5%	WB	48.6	49.4	52.0	56.5	63.2	69.9	73.9	74.5	69.6	64.1	56.9	52.0	(31)
(32)			MCD B	56.3	57.9	62.0	67.3	70.6	75.1	79.0	79.3	74.7	69.1	63.6	60.3	(32)
(33)		10%	WB	47.2	47.6	50.1	54.6	61.8	68.4	72.7	73.1	68.1	62.6	55.3	50.2	(33)
(34)			MCD B	54.1	55.4	59.4	63.4	68.7	73.9	77.9	78.2	73.3	67.5	61.5	57.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.2	9.2	9.9	10.5	10.2	11.0	11.3	11.2	11.2	9.8	9.8	9.2	(35)
(36)			MCDBR	13.2	15.0	16.4	17.6	15.5	13.8	12.7	12.4	13.6	13.4	13.9	14.4	(36)
(37)			MCWBR	6.9	7.5	7.8	8.1	8.7	8.9	9.0	8.9	9.4	7.9	7.4	7.7	(37)
(38)		5% WB	MCD BR	11.7	14.1	15.3	15.0	12.4	11.5	11.2	11.4	12.1	10.9	11.6	12.5	(38)
(39)			MCWBR	6.7	7.9	7.9	8.0	7.9	8.5	8.6	8.8	9.0	7.3	7.4	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.351	0.380	0.418	0.453	0.426	0.410	0.417	0.422	0.408	0.390	0.360	0.350	(40)
(41)		taud		2.357	2.269	2.178	2.112	2.270	2.361	2.356	2.338	2.360	2.374	2.403	2.383	(41)
(42)		Ebn at Noon		248	258	261	260	271	275	271	266	262	252	244	238	(42)
(43)		Edh at Noon		27	34	42	48	42	39	38	38	35	31	26	25	(43)
(44)	All-Sky Solar Radiation	RadAvg		455	698	1086	1514	1906	2176	2240	1962	1449	958	632	432	(44)
(45)		RadStd		60	99	92	156	165	164	94	130	112	85	80	59	(45)

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	+1.93	N/A	+1.46	+1.57	N/A	N/A	N/A
(47)	Regional (9 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page